

Modeling and Identification of Hysteresis in Robot Joints with Cycloidal Drives

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Abstract—Cycloidal drives (CDs) are precision transmissions that are often installed in the joints of industrial robots (IRs) with high payloads. It is well known that the hysteresis behavior of the CDs has a significant influence on the dynamic path accuracy of IRs. In contrast to harmonic drives (HDs), which are commonly used in lightweight robots, less literature has been published on the dynamic behavior of robot joints with CDs. Thus, this paper investigates the hysteresis behavior of robot joints with CDs. For the representation of the hysteresis behavior, a nonlinear autoregressive with exogenous inputs (NARX) model is proposed and compared to a Bouc-Wen model. Both, the data-driven NARX and the Bouc-Wen model are investigated experimentally in terms of their capability for representing the hysteresis behavior of CDs. The quality of the models is validated using a laboratory setup that mimics the drivetrain of a robot joint of the high payload class.

Index Terms—Industrial robot, flexible joint robot, Bouc-Wen model, NARX neural network

I. INTRODUCTION

In the 1980s, it was already discovered that the drivetrain of IRs has a significant influence on the obtainable dynamic path accuracy [1]. Whereas the drivetrain of IRs usually consists of a permanent magnet synchronous machine (PMSM) and a precision transmission [2]. Due to the high requirements regarding efficiency, accuracy, load and size, two types of transmissions - HD and CD - are commonly used. On account of the operating principle (rolling contact instead of teeth meshing), CDs are adopted for IRs with high payloads and HDs for lightweight robots. As a result, CDs are capable of withstanding higher torques while having a smaller size [3]. However, an additional attached planetary gear is required due to the smaller maximum gear ratio. Both, HDs and CDs, exhibit a nonlinear stiffness and friction behavior, which jointly yields a hysteresis characteristic.

Originating from the understanding of the influence of the drivetrain, the first modeling approaches concerning the drivetrain elasticity as a linear stiffness emerged [4]. Based on this, a variety of control approaches evolved [5]–[8]. However, the theoretical control performance of many of these approaches cannot be obtained experimentally for heavy IRs without significant modifications [9].

To further increase the accuracy and thus open up new

areas of application, more detailed models of the robot joints were developed. Most of these modeling approaches explicitly investigated the nonlinear dynamics of HDs with respect to friction and nonlinear stiffness. Pioneering work in the field of modeling HDs was performed by Seyffert et al. [10], Tuttle and Seering [11], Kircanski and Goldenberg [12], Taghirad and Bélanger [13] and Dhaouadi et al. [14], among others. In addition, the studies of Tjahjowidodo et al. [15] and Ruderman et al. [16]–[21], should be highlighted, as they also propose compensation approaches. Tjahjowidodo et al. [15] models the nonlinear dynamics using parallel Maxwell-slip elements. A comparison shows the superiority regarding the model proposed by Seyffert et al. [10], which represents the hysteresis by a weighted hyperbolic friction function and a Coulomb friction. In early works of Ruderman et al. [16], [17], the authors present a Preisach hysteresis model including a generalized Maxwell-Slip friction model. In recent work [20], a rate-independent Bouc-Wen hysteresis model is used in combination with a standard Stribeck friction model. The experimental validation is carried out for a single robot joint with a HD and an inertial disc as load. Besides its application in robotics, the Bouc-Wen model has also proven useful for modeling other hysteretic systems, such as piezoelectric actuators [22].

For piezoelectric actuators, recent modeling approaches [23] using NARX models based on a neural network show promising performance. It is known from the literature, that NARX models are capable of representing hysteresis behavior, especially Bouc-Wen models [24]–[26]. Moreover, it is known that NARX models can represent a wide range of nonlinear systems with high accuracy [27]. One of the main advantages of a NARX model is the easy extendability of the hysteresis model by further dependencies, such as temperature.

There is a variety of modeling approaches for the dynamic behavior of robot joints available. However, these approaches have been experimentally validated exclusively for HDs. Comparable experimental studies on the dynamic behavior of CDs have not yet been published. For HDs, the Bouc-Wen model has proven to be suitable for modeling the hysteresis behavior. In addition, NARX models represent another promising approach for modeling the hysteresis of robot joints. In this context, NARX models are particularly attractive concerning the extension of additional dependencies, such as temperature,

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as well as for potential compensation approaches.

In this paper, the dynamic behavior of an industrial robot joint with a CD is investigated. A Bouc-Wen as well as a NARX model based on a shallow neural network are identified using input-output measurements. Both, the Bouc-Wen and NARX models are experimentally validated and compared in terms of their ability to represent the hysteresis behavior.

The paper is organized as follows. Section II introduces both the Bouc-Wen and the NARX model for the representation of the hysteresis behavior of CDs. Subsequently, an experimental study of the dynamic behavior is carried out in Section III. Whereas the study involves the identification and validation of the models. Finally, concluding remarks are given in Section IV.

II. MODELING OF ROBOT JOINTS

The robot joint models proposed in this paper are based on the model structure given in Fig. 1. Assuming an ideal gear reduction, the joint position

$$q = u^{-1}\theta. \quad (1)$$

is given by the motor position θ and the ideal gear ratio u . In addition, the rolling contact inside of the CD leads to friction, which in combination with the nonlinear stiffness results in the previously mentioned hysteresis behavior. As a consequence, a torsion angle

$$\phi = u^{-1}\theta - q \quad (2)$$

between the motor θ and joint position q arises. The individual causes leading to hysteresis are almost impossible to decouple on the basis of input-output measurements. Due to this fact, the joint torque $\tau_g(q, \theta, t)$ is usually represented as a nonlinear function of the joint position q , the motor position θ and a time dependency. This nonlinear function can be represented by different modeling approaches. Depending on the modeling approach, a viscous damping D may be included as a function of the joint torsion velocity $\dot{\phi}$. However, from a practical point of view, the viscous damping can often be neglected [19].

Following [20], we use a classical Stribeck friction model to represent the motor-side friction

$$\tau_f = \left(F_c + (F_s - F_c) \cdot \exp \left(- \left| \frac{\dot{\theta}}{v_s} \right|^\delta \right) \right) \text{sign}(\dot{\theta}) + F_v \dot{\theta}, \quad (3)$$

with the Coulomb F_c , viscous F_v and static F_s friction coefficients. The parameters δ and v_s corresponds to the shaping factor and Stribeck velocity.

The coupling of rigid-body dynamics (gravity, inertial coupling, coriolis and the centrifugal terms) present in the case of a manipulator with multiple degrees of freedom as well as process forces are combined in the disturbance τ_{ext} . Finally, the dynamic model of a single robot joint

$$\ddot{q} = M^{-1} [\tau_g(q, \theta, t) + \tau_{\text{ext}}] \quad (4a)$$

$$\ddot{\theta} = J_m^{-1} [\tau_m - \tau_f - u^{-1}\tau_g(q, \theta, t)] \quad (4b)$$

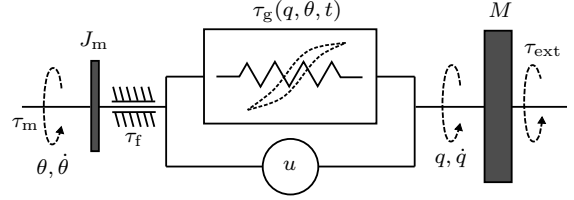


Fig. 1. General robot joint considering the hysteresis behavior (adapted from [19])

with the joint-side inertia M , the motor inertia J_m and the motor torque τ_m results.

A. Bouc-Wen model

The class of Bouc-Wen models [28], [29] originates from the modeling of nonlinear structural mechanical vibrations. Moreover, Bouc-Wen models have been proven to be suitable for modeling the hysteresis behavior of HD [19]. Therefore, we adopt the Bouc-Wen model of [19]

$$\tau_g(\phi, t) = wk(\phi(t))\phi(t) + (1 - w)k(\phi(t))x(t) \quad (5a)$$

$$\dot{x}(t) = \dot{\phi}(t) - \beta \left| \dot{\phi}(t) \right| |x(t)|^{n-1} x(t) - \gamma \dot{\phi}(t) |x(t)|^n, \quad (5b)$$

where the weighting factor $0 < w < 1$ reflects the relationship between purely plastic ($w = 0$) and purely elastic behavior ($w = 1$). This model is rate-independent, which implies that the hysteresis behavior is independent of the input frequency. For the nonlinear stiffness $k(\phi)$ a polynomial function

$$k(\phi) = k_0 + k_1|\phi| + k_2|\phi|^3 \quad (6)$$

is applied. The actual hysteresis behavior is captured with the state $x(t)$. Whereby, the shape and scale of the hysteresis is characterized by the parameters γ , β and n . It is known that some parameters of the model are redundant. Therefore, γ is often set to zero. For more details, we refer to the survey about Bouc-Wen models given in [30].

B. NARX model

NARX models have proven to be useful to describe a large class of nonlinear systems [27]. The general mathematical representation of a NARX model reads as follows:

$$\begin{aligned} \mathbf{y}(k+1) &= \mathbf{F}(\mathbf{y}(k), \mathbf{y}(k-1), \dots, \mathbf{y}(k-n_y+1), \\ &\quad \mathbf{u}(k), \mathbf{u}(k-1), \dots, \mathbf{u}(k-n_u+1)), \end{aligned} \quad (7)$$

where $\mathbf{F}(\cdot)$ represents a nonlinear vector function connecting the output $\mathbf{y} \in \mathbb{R}^{n_y \times N_y}$ and the input $\mathbf{u} \in \mathbb{R}^{n_u \times N_u}$ of the system. Both, the output $\mathbf{y}$ and the input $\mathbf{u}$ may include n delays as well as a N signals. Thus, the output at time k depends on the n_y past values of the output as well as the n_u past values of the input. The structure of a typical NARX model is given in Fig. 2.

In the following, the NARX model is used to represent the nonlinear function of the joint torque τ_g . For this purpose, an output-feedback recurrent neural network is implemented for the nonlinear function $\mathbf{F}(\cdot)$. It is well-known that such neural networks can efficiently approximate a large class of nonlinear

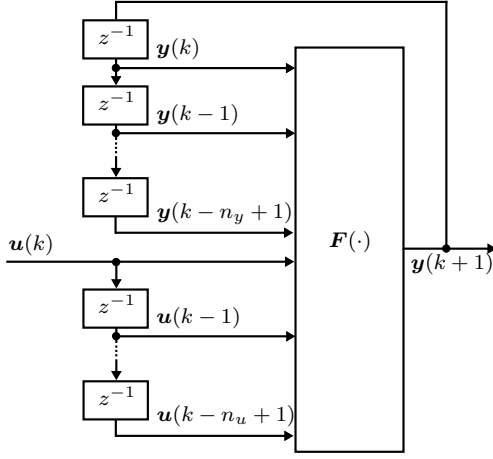


Fig. 2. Structure of NARX model (based on [31])

dynamical systems [31], [32].

One of the main tasks is the choice of hyperparameters that determine the network architecture (delays, hidden layer neurons). The output of each neuron o_j in the network is determined on the basis of a so-called activation function $a(\cdot)$ [32], as follows

$$o_j = a\left(\sum_i w_i x_i + b_i\right), \quad (8)$$

with the neurons inputs x_i , the weights w_i and the biases b_i . For the output layer neurons a pure linear

$$a(x) = x \quad (9)$$

and for the hidden layer a hyperbolic tangent sigmoid

$$a(x) = \text{tansig}(x_i) = \frac{2}{1 + \exp(-2x)} - 1 \quad (10)$$

activation function is applied. To avoid trial-and-error for the choice of the hyperparameter of the neural network, optimization methods can be used [33]. Therefore, Bayesian optimization [34] is applied to find the optimal number of neurons in the hidden layers. Whereas the delays are set based on an autocorrelation and cross-correlation analysis. The joint torsion velocity $\dot{\phi}$ is explicitly added as input to the neural network $F(\cdot)$, since it is known from [10], that this relation is a characteristic of systems with hysteresis behavior. Similarly, the velocity dependency is also present in the state x of the Bouc-Wen model (5). Consequently, the neural network does not need to reproduce this implicit dependency itself. Moreover, the motor velocity $\dot{\theta}$ is considered as additional input. The resulting neural network architecture $F(\cdot)$ with two hidden layers is shown in Fig. 3.

III. EXPERIMENTAL STUDY

The experimental study is carried out with a setup corresponding to a single robot joint. The aim of the case study is to investigate how the Bouc-Wen as well as the NARX model perform for representing the hysteresis behavior of CDs.

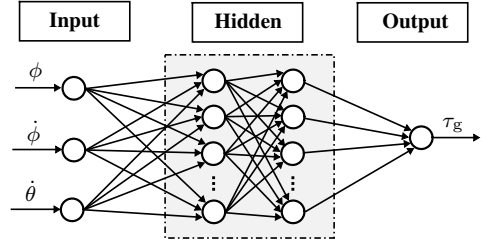


Fig. 3. Neural network $F(\cdot)$ architecture.

A. Experimental Setup

The experimental setup, which is shown in Fig. 4, mimics the drivetrain of a robot joint with one rotational degree of freedom. As precision transmission the *RH380-N* CD from Nabtesco ① is applied. This transmission has a nominal gear ratio u of 185, a rated torque of 3.7 kNm and a maximum acceleration torque of 9.3 kNm, which is equivalent to the heavy IR payload class. The CD is actuated on the motor-side with the *MSK070D* PMSM from Bosch Rexroth ② with a maximum torque of 52.5 Nm and an input encoder with a resolution of 13 Bit. Both, the joint position as well as the joint torque are measured. For the position measurement the high-precision rotary encoder *ECN 1325* of Heidenhain with a resolution of 25 Bit is used. For the measurement of the joint torque the *T40B* torque sensor of HBM ③ is used in combination with the analog input clamp *EL3102* of Beckhoff with a resolution of 16 Bit. The torque sensor has a rated torque of 10 kNm and is specifically designed for test bench applications in research and end-of-line production testing. The experimental setup has the special feature that the transmission is not simply loaded by an inertial disc or brake. In contrast, a dynamic load torque can be applied via a joint-side motor. Therefore, the output shaft is connected to a high-torque motor using the *Roba DS 1400* double-jointed coupling of Mayr ④ with a torsional stiffness of about 15 kNm/rad. The water-cooled high-torque motor is the *DST2-315KO* of Baumüller ⑤ with a maximum torque of 3.4 kNm. This provides the possibility to simulate the coupling torques of a complete manipulator as well as the impact of process forces. Furthermore, the hysteresis behavior can be investigated dynamically under reproducible conditions. The current control of both motors is executed on industrial motion controllers of Bosch Rexroth and Baumüller, respec-

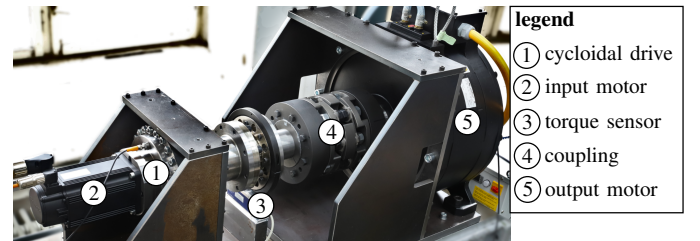


Fig. 4. Test bench with single CD robot joint for the experimental study.

tively. The desired trajectory generation as well as the cascaded P-Pi control of the input motor is carried out with rapid prototyping hardware of Speedgoat, which is programmed with Simulink Real-Time. The motion controllers and the additional measurement hardware communicate with the rapid prototyping system via a EtherCAT bus with a cycle time of 4 kHz.

B. Identification

The system identification is carried out using input-output measurements, whereby the focus of identification lies on the hysteresis behavior of the joint torque τ_g . Therefore, it is assumed that the motor inertia J_m and joint inertia M can be obtained from CAD data. The identification approaches proposed below require the measurement of the joint torque τ_g as well as the motor θ and joint position q . Whereas common industrial manipulators only have sensors on the motor-side for cost reasons. However, CDs are usually tested under static loads as part of an end-of-line inspection, where sensors for the joint torque τ_g and joint position q are available. Theoretically, this inspection can be extended to include an identification procedure. The parameters determined in this way can then be stored in the robot controller.

The identification of Bouc-Wen model (5) is due to its nonlinear and non-differentiable behavior a challenging problem [30]. As a consequence, many identification techniques have been applied, among those particle swarm optimization (PSO) has proven to be useful [35]. Therefore, in this study, PSO with the following parameters is used: Swarm size of 300 particles, cognitive parameter $c_1 = 1.49$, social parameter $c_2 = 1.49$ and inertia range between 0.1 and 1.1. A discussion of the influence of this basic PSO parameters is provided in [36]. As experiment, a triangle excitation signal of the torque of the joint-side motor τ_{ext} with a frequency of 2 Hz with increasing amplitude from 1 kNm to 3 kNm is applied. Subsequently, the PSO is used to minimize the mean squared error (MSE) between the torque sensor signal and the model output. The implementation is realized with the Global Optimization Toolbox of MATLAB. The identified parameters of the Bouc-Wen model are listed in Table I.

For the identification of the NARX model, the MSE is minimized by adjusting the weightings and biases of the neural network. During the training, the true output (joint torque τ_g)

TABLE I
IDENTIFIED BOUC-WEN PARAMETERS

Parameter	Description	Value	Unit
k_0	Stiffness coefficient	1.124e6	Nm
k_1	Stiffness coefficient	9.216e7	Nm/rad
k_2	Stiffness coefficient	9.964e7	Nm/rad ³
w	weighting factor	0.885	—
β	shaping factor	8.871e6	—
γ	shaping factor	0	—
n	shaping factor	2.037	—

is available by means of the input-output measurements. Therefore, the neural network is trained in series-parallel architecture [31]. After successful training, the neural network is rearranged to parallel architecture. This approach avoids the existence of feedback loops during the training. Hence, the training is performed using Bayesian regularization backpropagation. The neural network with two hidden layers is evaluated at a sample time of 2 kHz. Using autocorrelation and cross-correlation analysis, delays of 8, 12, 1 and 3 are determined for the joint torsion ϕ , joint torsion velocity $\dot{\phi}$, motor velocity $\dot{\theta}$ and the output $\tau_{g,NARX}$, respectively. Moreover, Bayesian optimization is used to determine the number of neurons in the hidden layers to 18 and 8. The training data consists of a triangle signal and a random binary signal of motor position as well as of the load torque. The implementation and training of the NARX model is done with the Deep Learning Toolbox of MATLAB [37].

Subsequently, the Stribeck friction model (3) is identified using closed-loop motion trajectories with constant velocity. Therefore, several point-to-point movements are performed between which the absolute value of the constant velocity is increased from $0.01u\text{ rad/s}$ to $0.7u\text{ rad/s}$. According to the dynamics model (4), the friction torque τ_f in case of a constant velocity is approximately given by $\tau_f \approx \tau_m - u^{-1}\tau_g$. The constant velocity areas are identified and the joint torque is subtracted from the motor torque. The preprocessed input-output measurements are subsequently used to identify the friction parameters by minimizing the MSE with the Levenberg-Marquardt algorithm. The identified Stribeck friction coefficients are listed in Table II. Furthermore, the measured and identified friction curves are shown in Fig. 5.

C. Validation

In this subsection, the identified models are validated in terms of their ability to capture the hysteretic behavior of

TABLE II
IDENTIFIED STRIBECK FRICTION PARAMETERS

Parameter	Description	Value	Unit
F_s	Static coefficient	2.13	Nm
F_c	Coulomb coefficient	1.63	Nm
F_v	Viscous coefficient	0.0190	Nms/rad
v_s	Stribeck velocity	30.2	rad/s
δ	Shaping factor	1.82	—

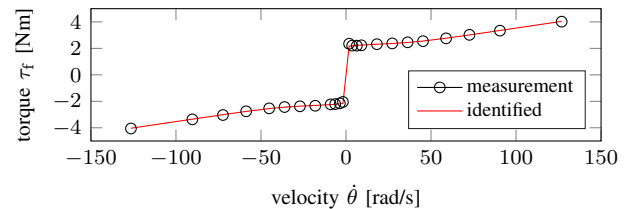


Fig. 5. Measured and identified Stribeck friction of the CD robot joint.

CDs. For this purpose, a static hysteresis curve is generated by plotting the joint torque τ_g versus the torsion angle ϕ . The curves, as shown in Fig. 6, are obtained by exciting the system with a sine signal of the torque of the joint-side motor τ_{ext} with an amplitude of 3 kNm and a frequency of 2 Hz. Both, the Bouc-Wen as well as the NARX model are able to represent this relationship. Whereas, the Bouc-Wen model achieves a higher accuracy in the range of small joint torsion and the NARX model in the range of larger torsion. Furthermore, using the identical sinusoidal load signal, further hysteresis curves are determined, whereby meanwhile the input motor is operated at constant velocity $\dot{\theta}$ of $0.2u \text{ rad/s}$ in both directions of rotation. Depending on the motor velocity, this leads to an additional torsion, which causes a shift of the hysteresis curve (cf. Fig. 7). It should be noted that different encoders are used to measure the motor and joint position. The encoder signals and their signal processing may have different delays, which can result in the measurement of an additional torsion, as observed in experiment. However, the basic shape of the hysteresis curve is preserved.

Moreover, a robotics application scenario is simulated by means of multiple trapezoidal point-to-point trajectories with a maximum velocity of 0.4 rad/s . Simultaneously, a load torque τ_{ext} is applied as sawtooth signal with a maximum amplitude of 3 kNm using the joint-side motor, which imitates a machining task. Figure 8 shows the corresponding desired joint position q_d , motor velocity including the gear ratio $u^{-1}\dot{\theta}$, joint torsion ϕ , along with the desired load $\tau_{\text{ext},d}$ and joint torque τ_g of the trajectory tracking experiment. Moreover, the estimated joint torque using the Bouc-Wen $\tau_{g,BW}$ as well as the NARX model $\tau_{g,NARX}$ are shown. As soon as the motor velocity $\dot{\theta}$ is approximately zero, the estimated torque of the Bouc-Wen model $\tau_{g,BW}$ coincides with the measured one τ_g . The NARX model, in comparison to the Bouc-Wen model, performs better in the phases with non-zero motor velocity. Due to the additional input of the motor velocity, the NARX model can reproduce the dependency on the motor velocity. The error between measured torque signal and estimated joint torques during the trajectory tracking experiment is shown

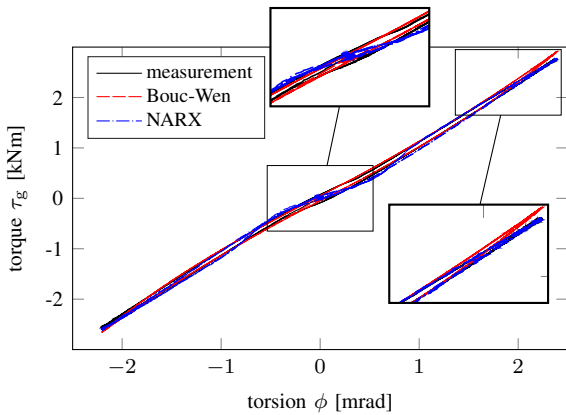


Fig. 6. Measured and identified hysteresis of the CD robot joint.

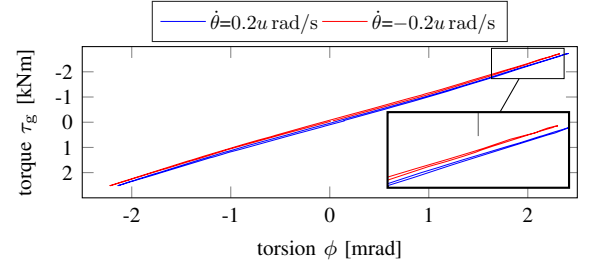


Fig. 7. Measured hysteresis of the CD robot joint with constant motor velocity $\dot{\theta}$.

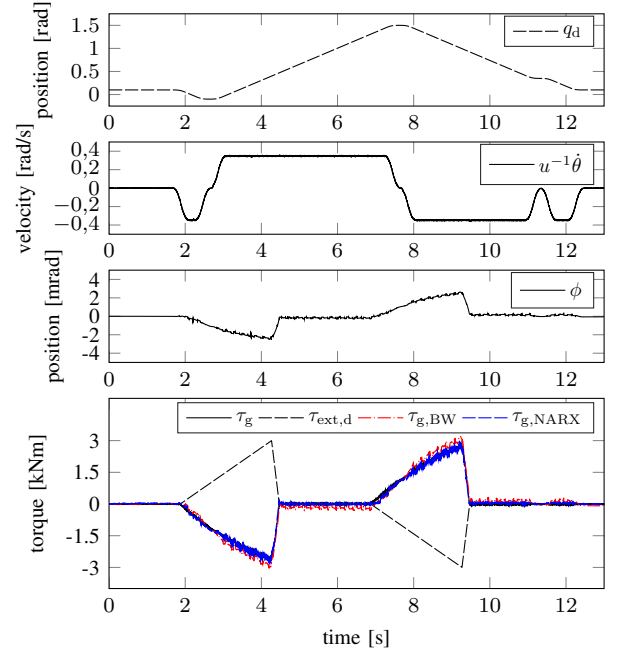


Fig. 8. Trajectory tracking experiment to evaluate hysteresis models.

in Fig. 9. The RMSE of the estimated joint torque of the NARX model is reduced by 75 % compared to the Bouc-Wen model. The maximum error is reduced by 35 %. However, the accuracy improvement of the NARX model comes at the cost of an increased identification effort due to the larger number of model parameters.

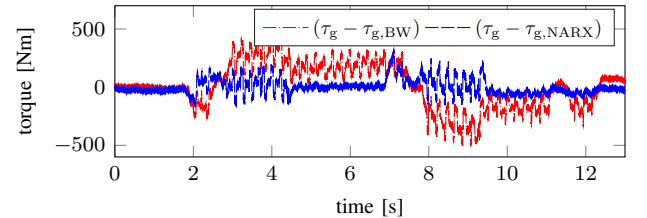


Fig. 9. Error between measured torque signal and estimated joint torques during the trajectory tracking experiment (cf. Fig. 8 bottom).

IV. CONCLUSION

In this paper, a modeling approach of the hysteresis behavior of robot joints with CDs was proposed. The presented experimental results demonstrate that both the Bouc-Wen as well as the proposed NARX model are capable of representing the static hysteresis behavior. The experimental study has further revealed a motor velocity dependent joint torsion. Thereby, one advantage of the NARX model is that it can be extended by such additional dependencies. Consequently, using the NARX model, a higher accuracy is achieved.

Current research concerns the investigation of further dependencies of the hysteresis behavior, such as temperature. In addition, we are focusing on the extension of the Bouc-Wen model by the motor velocity dependency, since a white-box approach is more attractive from the point of view of analysis and industrial acceptance. Furthermore, we aim to use the models for a sensorless compensation of the hysteresis.

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